11.4 Modeling with Area

Name _____ Period ____ Date _____

Geometry 11.4 (30-minute plan) · Student sheet

I can…

- find and compare population densities.
- find areas of real-life regions and convert between square units.
- find the rectangle with the greatest area for a given perimeter.

Vocabulary · fill in the missing word

Population density — a _____ of how many people live within a given area

Square unit — the _____ of a square with sides of one unit

Part 1 · Notes and examples (fill in each note, then work the example)

Population density = _____ ÷ _____

Example 1

About 2,240 people live in a town that covers 4.5 square miles. Find the population density. Round to the nearest tenth.

Area first: when a region is a circle or polygon, find its _____ with a formula, then divide.

Example 2

About 52,000 people live within 6 miles of a city's center. Find the population density of that circular region. Round to the nearest tenth.

Square units: 1 yd^2 = _____ ft^2 (3 ft × 3 ft), so divide square feet by 9 to get square yards.

Greatest area: for a fixed perimeter, the rectangle with the greatest area is a _____.

Example 3

A lawn has an area of 1,800 square feet. Sod costs $3.50 per square yard. How much will the sod cost?

Part 2 · Partner practice

Partner practice

A circular park with a radius of 0.4 mile has about 1,500 trees. Find the number of trees per square mile. Round to the nearest whole number.